



Typical solution

Course-related questions (4 pts)

- 1) We can't use *scanf* to read a String value containing spaces. (1.5 pt)
- 2) In algorithmics, is there any function that allows resizing a static array? (Yes / No) (0.5 pt)
- 3) Replace the loop "For" in the following algorithm by the loop "While"

```
i=1;
while (i <= 10) do
    Instructions
    i = i + 2;
End-while
```

Exercise 1 (5.5)

```
#include<stdio.h>
int n, i;
main(){
    do {
        printf("Give the number of values : ");
1       scanf("%d", &n);
    } while (n > 10) ;
2       int t[n];
3       for (i=0; i < n; i++){
4           printf("Give value n° %d : ", i+1);
5           scanf("%d", &t[i]);
        }
6       for (i=n-1; i >= 2; i--){
7           t[i+1] = t[i];
        }
8       t[2] = 14;
    }
```

1) 0.25 / row (only 10 first rows)

	n	i				
(1)	4	?				
(2)	4		T[0]	T[1]	T[2]	T[3]
(3)	4	0	?	?	?	?
(5)	4	0	8	?	?	?
(3)	4	1	8	?	?	?
(5)	4	1	8	-2	?	?
(3)	4	2	8	-2	?	?
(5)	4	2	8	-2	3	?
(3)	4	3	8	-2	3	?
(5)	4	3	8	-2	3	7
(6)	4	3	8	-2	3	7
(7)						

- 2) The problem is : T[4] does not exist. 0.5 pt
- 3) To fix this problem, we have to declare the array T before main, with the maximum size 11. 1 pt

The other solutions like replace T[n] by T[n+1] 0.5 pt

- 4)
 - a) Read the size of the array. The max size is 10. 0.25 pt + 0.25 pt
 - b) Read the elements of the array, 0.25 pt
 - c) Right shift of elements of the array starting from the third element to the last element. 0.5 pt
 - d) Assign the value 14 to the third position of T 0.25



Exercise 2 (6 pts)

```
#include <stdio.h>
float num1, num2;
char operator;
```

General form + correct declaration : 1.75 pt

-0.25 for each missing element

```
main() {
    printf("Enter the first number: ");
    scanf("%f", &num1);

    printf("Enter an operator (+, -, *, /): ");
    scanf(" %c", &operator);

    printf("Enter the second number: ");
    scanf("%f", &num2);

    // Perform the operation based on the operator
    switch (operator) {
        case '+':
            printf("Result: %f + %f = %f\n", num1, num2, num1 + num2);
            break;
        case '-':
            printf("Result: %f - %f = %f\n", num1, num2, num1 - num2);
            break;
        case '*':
            printf("Result: %f * %f = %f\n", num1, num2, num1 * num2);
            break;
        case '/':
            if (num2 != 0) {
                printf("Result: %f / %f = %f\n", num1, num2, num1 / num2);
            } else {
                printf("Error: Division by zero is not allowed.\n");
            }
            break;
        default:
            printf("Error: Invalid operator. Please use +, -, *, or /\n");
    }
}
```

0.5 pt

0.5 pt

0.5 pt

0.5 pt

0.5 pt

0.5 pt

1 pt

0.25 pt



Exercise 3 (5 pts)

Algorithm SommeE ;

Var i,N,S : Integer ;

E:real;

Begin

Repeat

Read(N) ;0.5

Until N > 2;0.5

E ← 14 ; S ← 1;0.5

For i ranging from 2 to N do0.5

S ← S+i;0.5

E ← E+1/S ;0.5

End-for ;

Write('E=',E) ;0.5

End.

General Form of the algorithm (Algorithm, Begin,
End, Repeat, ...) and correct declarations : 1.5 pt
(-0.25/error)

La délimitation d'un bloc par
les mots clés Begin-End est
aussi acceptée. Exemple :
For i ranging from 2 to N do
Begin
End;